Digital Signal Processing Theory

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Digital Signal Processing Theory (A11846)

Short Title:	DSP Theory		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Electronics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

Digital Signal Processing (DSP) techniques play a major role in many technological areas. DSP is a mathematical subject which is concerned with the representation and transformation of discrete-time signals using digital computation. The module focuses on the fundamentals of DSP and explores speech processing as a paradigm for DSP application techniques.

Programmes

	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	4 / 7 / M
`ENGR-0020`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	4 / 7 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	4 / 7 / M
`ENGR-0020`	BSc (Hons) in Applied Electronics (`WD_EELEC_B`)	2 / 3 / M
`ENGR-0020`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	4 / 7 / E

Indicative Content

- Discrete-time signals and properties
- Discrete-time LTI Systems and properties
- Transforms: Z, DtFT, DFT, their relationships and use in the frequency domain analysis of LTI systems
- Goertzel algorithm (DTMF tone detection)
- Window functions, FIR and IIR filter design
- Speech modelling and time domain analysis of speech signals
- Frequency domain analysis of speech signals: Short-time Fourier Transform (STFT)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate knowledge and understanding of discrete-time signals and LTI systems.
2. Analyse discrete-time systems to determine system behaviour in the time and frequency domain.
3. Formulate and manipulate equations describing systems and transforms.
4. Design digital filters (FIR and IIR) from given specifications.
5. Apply Digital Signal Processing techniques to non-stationary signals.
6. Write simulation programs to analyse, interpret and evaluate discrete-time systems behaviour.
7. Communicate effectively, information of experimental nature with due regard to the interpretation of results and the underpinning theory.

Learning and Teaching Methods

- Lectures and tutorials
- Computer-based laboratory work

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1, 2, 3, 4, 5
Final Practical	40%	6, 7

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Proakis, J.G. and D.G. Manolakis. _Digital Signal Processing: principles, algorithms, and applications_. 4th ed.. New Jersey: Prentice Hall, 2006.

Supplementary Material

- Mitra, S.K. _Digital Signal Processing: a computer based approach_. 3rd ed.. New York: McGraw Hill, 2006.

Requested Resources

- ENGINEERING LAB: Electronics